

OPCION 2:

Calculated Parameters		Unit	Side 1	Side 2
Flow Type			CounterCurrent	
Heat Load		BTU/h	4878051.99	
Inlet Temperature		°F	90.0	78.0
Outlet Temperature		°F	80.0	88.0
Mass Flow Rate		lb/h	488970.56	488950.68
Volumetric Flow Rate		GPM (US)	980.00	979.65
Total Pressure Drop		psi(g)	8.65	8.65
Pressure Drop in Port		psi(g)	0.98	0.98
Fouling Factor		Hrft ² °F/KBTU	0.0414	0.0414
Surface Margin		%	10.04	
LMTD		°F	2.0	
HTC (Available/Required)		BTU/ft ² ·hr·°F	1212 / 1102	
Port Velocity		ft/s	11.48	11.47
Shear Stress		Pa	58.65	58.63
Properties of Fluid		Unit	Side 1	Side 2
Fluid			Water	Water
Liquid Viscosity		cP	0.8090	0.8336
Liquid Density		lb/ft ³	62.2042	62.2287
Liquid Heat Capacity		BTU/lb·°F	0.9976	0.9977
Liquid Thermal Conductivity		BTU/h·ft·°F	0.3536	0.3523
Specifications		Unit	Side 1	Side 2
HEX Type			S62-DS10-304-TMTL92	
Number of Plates			304	
Grouping			1x152 + 0x0 / 1x151 + 0x0	
Plate Thickness		in	0.016	
Plate Material / Ratio			AISI304 / 47%	
Effective Area		ft ²	2213.73	
Gasket Material			NBRH FDA (SonderLock) 5/302 °F	
Frame	Type		DS, painted frame	
	Length	ft	8.48	
	Maximum Number of Plates		395	
Volume		ft ³	11.3	11.2
Weight, empty/operating		lb	3477 / 4875	
Paint Category			Category C2L	
Paint Color			TRAFFIC RED RAL 3020	
Connection	Inlet		F1: 6" B16.5#150 Studded end, 316 clad	F3: 6" B16.5#150 Studded end, 316 clad
	Outlet		F4: 6" B16.5#150 Studded end, 316 clad	F2: 6" B16.5#150 Studded end, 316 clad
Certification/Approval Type			None	
Minimum Design Temperature		°F	78.0	
Maximum Design Temperature		°F	90.0	
Maximum Differential Pressure		psi(g)	145.0	
Maximum Test Pressure		psi(g)	188.5	
Maximum Design Pressure		psi(g)	145.0	145.0

Notes:

Aluminum shroud omitted

The purchaser of the equipment bears total responsibility for suitability of use of all materials in this application.

We may have assumed some design values. If they differ from your requirements, a new design may be necessary.

THE HEAT AND MASS TRANSFER EXPERTS
 PLATE & DYNAMIC SCRAPPED SURFACE HEAT EXCHANGER
 CORRUGATED TUBULAR HEAT EXCHANGER - PISTON PUMPS
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